AIM-Fair: Advancing Algorithmic Fairness via Selectively Fine-Tuning Biased Models with Contextual Synthetic Data

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Abstract

Facial attribute classifiers trained on real-world data often inherit spurious correlations with demographic attributes, causing large worst-group performance gaps even when average accuracy remains high. Existing mitigation methods either rely on protected-group annotations, directly edit real images, or use generated samples without explicitly controlling the parameters that should adapt to synthetic data. We present AIM-Fair, a two-stage framework that improves group fairness by combining contextual synthetic data generation with selective fine-tuning. First, we construct balanced synthetic images using GPT-4-authored contextual prompts and a latent diffusion model, which increases attribute diversity beyond plain prompts. Second, we rank model parameters with gradient differences computed on biased real data, biased synthetic data, and balanced synthetic data, then fine-tune only parameters that are fairness-sensitive but relatively domain-insensitive. On CelebA, AIM-Fairimproves the Smiling/Male setting from DiGA's 81.90 worst-group accuracy and 7.40 equalized odds to 84.20 and 6.07, and improves Smiling/Young from 78.50 and 9.50 to 87.78 and 5.44. On the harder Young/Male setting, AIM-Fairtrades some overall accuracy for substantially better worst-group accuracy and equalized odds than the previous best generative baseline. On UTKFace, it also outperforms full fine-tuning in worst-group accuracy and equalized odds. These results show that fairness gains from synthetic data depend not only on what data are generated, but also on which pretrained parameters are allowed to move.

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1. Introduction

Machine learning systems deployed in socially sensitive settings frequently inherit demographic biases from their training data, producing uneven error rates across groups even when aggregate accuracy appears strong [14, 21]. This problem is especially visible in facial attribute classification, where target labels such as *Smiling* or *Young* are often spuriously correlated with protected attributes such as gender or age in the training set [12, 16, 24]. In practice, the most harmed demographic groups are typically minorities in the data distribution, so improving worst-group performance is at least as important as improving the average.

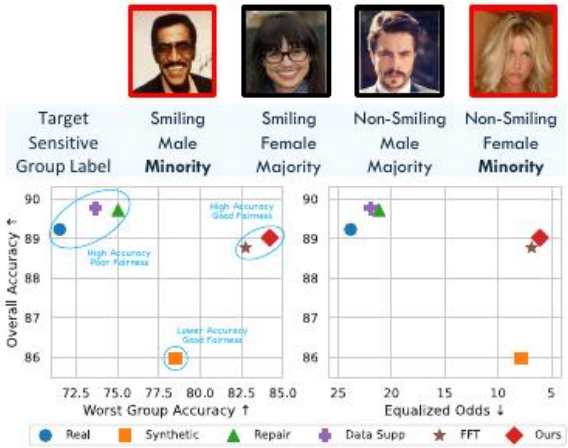

Figure 1. Teaser on CelebA with target attribute *Smiling* and protected attribute *Male*. Real-data training yields high utility but poor fairness; training only on synthetic data improves fairness but sacrifices accuracy; repairing or supplementing data remains suboptimal. AIM-Fairmoves the operating point toward better worst-group accuracy and lower equalized odds while preserving competitive overall accuracy.

Recent work has explored robust optimization, representation learning, and data generation for mitigating such biases [4, 10, 11, 13, 15, 25]. Among these, synthetic data are appealing because they can expose the model to demographic combinations that are rare in the observed training set. However, existing generative approaches still face two coupled problems. First, low-information prompts or direct data supplementation often produce images that do not sufficiently close the gap between synthetic and real domains [19, 22]. Second, even when synthetic data are useful, fully fine-tuning a pretrained classifier can overwrite task-relevant structure learned from real data, while generic partial fine-tuning does not explicitly target fairness.

This paper studies a simple question: can synthetic data improve fairness more reliably if we update only the parameters that react to fairness cues, while leaving domain-sensitive parameters largely fixed? Our answer is AIM-Fair, a framework that combines *contextual synthetic data generation* (CSDG) with *selective fine-tuning* (SFT). CSDG uses a bank of contextual prompts, authored with GPT-4 and instantiated with a latent diffusion model [18], to generate balanced facial images that vary in facial features, hairstyle, accessories, and head pose. SFT then uses three datasets, biased real, biased synthetic, and balanced synthetic, to estimate per-parameter gradient differences. Intuitively, parameters whose gradients change strongly between biased and balanced synthetic data are promising fairness carriers, whereas parameters whose gradients change strongly between real and synthetic data are more likely to encode domain shift. AIM-Fairupdates only the intersection of fairness-sensitive and domain-insensitive parameters.

The empirical pattern is consistent across the provided experiments. On CelebA, AIM-Fairimproves both worst-group accuracy and equalized odds over prior group-robust baselines and over the recent generative method DiGA [24]; on UTKFace it also improves over full fine-tuning. At the same time, the results remain calibrated: on the hardest Young/Male setting, AIM-Fairimproves fairness substantially but does not achieve the best average accuracy, showing that stronger fairness can still involve a utility trade-off.

Our contributions are threefold. First, we introduce a synthetic-data-based fairness adaptation pipeline that does not use protected attributes of real images during fine-tuning or mask generation, relying instead on controlled synthetic data. Second, we propose a parameter ranking rule based on absolute gradient differences across real and synthetic environments. Third, we show strong fairness-utility trade-offs across CelebA and UTKFace, together with ablations that identify when prompt diversity, bias ratio, synthetic-data amount, and mask size matter.

2. Related Work

2.1. Fair Visual Recognition and Group-Robust Learning

Bias mitigation in visual recognition has been studied through robust optimization, adversarial or disentangled representations, and training strategies that target spurious correlations [3, 13, 16, 21]. Group-robust baselines such as CVaR DRO [10], environment inference [4], learning-from-failure [15], JTT [11], and sample reweighting [25] improve worst-group performance without always requiring group labels at every training point, but they do not explicitly exploit synthetic counterfactual data. In facial attribute classification, fairness-aware objectives can be effective [16], yet they still operate within the support of the observed data and may struggle when minority attribute combinations are underrepresented.

2.2. Fairness with Limited or Missing Demographic Labels

The practical difficulty of collecting protected-attribute annotations has motivated work on fairness under partial supervision or without demographic data [1, 8]. These methods improve realism relative to fully supervised fairness setups, but they often rely on pseudo-label quality, external annotators, or specialized fairness regularizers. Our setting is different: the biased benchmark splits still define protected attributes for evaluation, but the adaptation stage of AIM-Fairdoes not require protected labels from real training images. Instead, we move the fairness burden to synthetic data whose target and protected attributes are explicitly controlled at generation time.

2.3. Synthetic Data for Bias Mitigation

Generative approaches have become increasingly prominent for fairness. Earlier work uses GAN-based latent editing to produce balanced augmentations [17], while more recent methods use diffusion or vision-language guidance to manipulate protected characteristics [5, 24]. More broadly, fair generative modeling and inclusive text-to-image generation aim to reduce representation bias at the generator level [2, 20, 23]. Our approach is complementary to these efforts. Rather than requiring a generator alone to remove all bias, AIM-Fairuses synthetic images as a controlled intervention signal and focuses on *which classifier parameters* should be updated. This selective adaptation perspective is also distinct from generic parameter-efficient fine-tuning [7, 9], because our parameter mask is driven by fairness and domain criteria rather than architecture alone.

3. Method

3.1. Problem Setup

We consider binary facial attribute classification with target label $y \in \{0, 1\}$ and protected attribute $s \in \{0, 1\}$. A pretrained classifier f_θ is first obtained by standard supervised training on a biased real dataset $\mathcal{D}_R$, where the target label is available and the empirical distribution exhibits spurious correlation between y and s. The goal is to improve worst-group performance and equalized odds after adaptation, while preserving strong average accuracy.

Following the released implementation, the backbone is a pretrained ResNet-18 [6] with a single binary output head. Evaluation uses four group accuracies $\mathrm{Acc}(y, s)$, overall accuracy, worst-group accuracy $\min_{y,s} \mathrm{Acc}(y, s)$, equalized odds, and the standard deviation of group accuracies.

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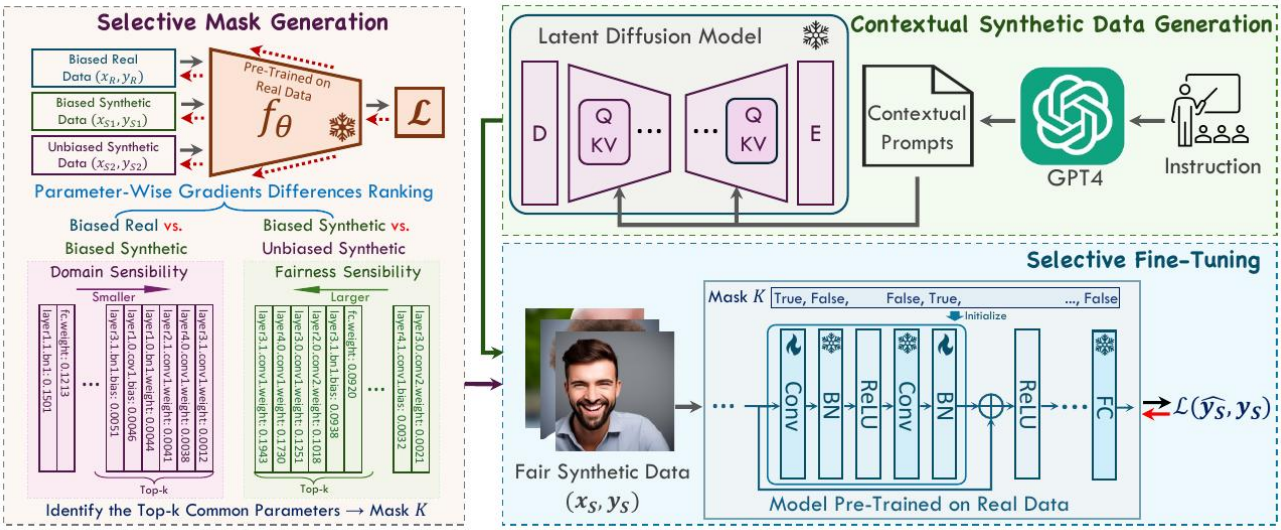

Figure 2. Overview of AIM-Fair. Left: selective mask generation ranks parameters by domain sensitivity and fairness sensitivity using average gradient differences computed on biased real, biased synthetic, and balanced synthetic data. Right: contextual synthetic data generation creates controlled balanced samples, and selective fine-tuning updates only the parameters in the resulting mask.

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3.2. Contextual Synthetic Data Generation

The first component of AIM-Fairis CSDG. Instead of using a single template prompt such as "portrait face photo of a smiling male," we construct contextual prompts that preserve the target and protected attributes while varying secondary factors such as facial features, hair characteristics, accessories, and head pose. The provided prompt bank contains 50 contextual prompts per attribute combination, with examples such as variations in cheekbones, eyeglasses, hats, lipstick, and viewing angle. These prompts are used to drive a latent diffusion model [18]; in the project assets, the generation scripts are based on Stable Diffusion v2.1.

This design addresses a concrete failure mode of plain prompting: if generated images are too homogeneous, synthetic balancing does not expose the classifier to enough within-group variation. The prompt ablation in Table 5 and qualitative samples in Figure 5 confirm that richer prompts produce more diverse, task-relevant synthetic images than a single plain description.

From these prompts, we build two synthetic datasets. The balanced synthetic set $\mathcal{D}_{S2}$ contains all four (y, s) combinations with equal sampling. The biased synthetic set $\mathcal{D}_{S1}$ matches the skewed real-data correlation more closely. In the main CelebA setting, the code uses 5,000 images per group for $\mathcal{D}_{S2}$ and a 1:9 minority-to-majority ratio for $\mathcal{D}_{S1}$.

3.3. Selective Mask Generation

The second component is selective mask generation. Starting from the real-data pretrained model f_θ, we compute parameter-wise average gradients on the three datasets:

$$g_R(p) = \nabla_{\theta_p} \mathcal{L}(\mathcal{D}_R), \tag{1}$$

$$g_{S1}(p) = \nabla_{\theta_p} \mathcal{L}(\mathcal{D}_{S1}), \tag{2}$$

$$g_{S2}(p) = \nabla_{\theta_p} \mathcal{L}(\mathcal{D}_{S2}). \tag{3}$$

where θ_p denotes a parameter tensor and $\mathcal{L}$ is the binary cross-entropy loss.

We then score each parameter tensor with two absolute-difference criteria:

$$d_{\text{domain}}(p) = \text{mean}(|g_R(p) - g_{S1}(p)|)\,, \tag{4}$$

$$d_{\text{fair}}(p) = \text{mean}(|g_{S1}(p) - g_{S2}(p)|)\,. \tag{5}$$

The first score estimates domain sensitivity: parameters with small d_{domain} respond similarly to real and synthetic biased data and are therefore safer to update with synthetic supervision. The second score estimates fairness sensitivity: parameters with large d_{fair} react strongly when the synthetic data distribution changes from biased to balanced, suggesting that they are relevant to debiasing.

Let $\mathcal{T}^k_{\text{domain}}$ be the set of k parameter tensors with the *smallest* d_{domain}, and $\mathcal{T}^k_{\text{fair}}$ the set of k tensors with the *largest* d_{fair}. The update mask is

$$\mathcal{K} = \mathcal{T}^k_{\text{domain}} \cap \mathcal{T}^k_{\text{fair}}. \tag{6}$$

All tensors outside $\mathcal{K}$ are frozen during adaptation. In the provided training script, the best-performing setting uses $k = 55$.

We rank parameter tensors rather than scalar weights because tensor-level masking matches optimizer granularity

and avoids fragmented updates that are difficult to reproduce. In practice, average gradients are accumulated separately on $\mathcal{D}_R$, $\mathcal{D}_{S1}$, and $\mathcal{D}_{S2}$ before selective fine-tuning. Mask construction is therefore an offline preprocessing step rather than an additional objective optimized jointly with training.

3.4. Selective Fine-Tuning

After mask generation, AIM-Fairfine-tunes only the selected parameters on the balanced synthetic set $\mathcal{D}_{S2}$:

$$\min_{\theta_{\mathcal{K}}} \mathbb{E}_{(x,y)\sim\mathcal{D}_{S2}} \left[\ell(f_\theta(x), y)\right], \tag{7}$$

while keeping $\theta_{\setminus\mathcal{K}}$ fixed. This procedure can be interpreted as a constrained adaptation step. The pretrained model retains most of the utility learned from real data, while the unfrozen subset absorbs fairness-relevant updates induced by the balanced synthetic data.

Unlike generic block freezing or random partial tuning, the mask is not chosen by architecture heuristics. Unlike full fine-tuning, the method explicitly resists synthetic-to-real domain drift. And unlike synthetic-only training, it starts from a classifier already aligned with the real domain.

4. Experiments

4.1. Experimental Setup

We evaluate AIM-Fairon CelebA [12] and UTKFace. CelebA includes three target/protected settings from the provided tables: *Smiling/Male*, *Smiling/Young*, and the harder *Young/Male*. UTKFace uses *Female* as the target and *White* as the protected attribute. The released code and scripts show a ResNet-18 backbone pretrained on ImageNet, SGD optimization, binary cross-entropy loss, image resolution 224×224, 15 epochs of pretraining on biased real data, and 10 epochs of masked fine-tuning on balanced synthetic data. The main CelebA setting uses $20{,}000$ biased real training images, balanced synthetic data with $5{,}000$ images per group, an imbalanced synthetic ratio of 1:9, and ten random seeds.

We compare with standard ERM pretraining, robust/group-aware baselines reported in the provided tables, the recent generative method DiGA [24], and several adaptation strategies including linear probing, full fine-tuning, synthetic-only training, data supplementation, data repairing, random parameter selection, SubTuning [9], and cosine-similarity-based mask selection.

4.2. Main Results

Table 1 shows that AIM-Fairconsistently improves the fairness-utility trade-off on CelebA. On *Smiling/Male*, it reaches 89.02 accuracy, 84.20 worst-group accuracy, and 6.07 equalized odds, surpassing DiGA's 88.40, 81.90, and 7.40. The gains are larger on *Smiling/Young*, where AIM-Fairimproves worst-group accuracy from 78.50 to 87.78 and reduces equalized odds from 9.50 to 5.44, while also increasing average accuracy from 89.10 to 90.21. On *Young/Male*, the pattern is more nuanced: DiGA retains the best average accuracy at 80.00, whereas AIM-Fairimproves worst-group accuracy from 51.30 to 54.89 and reduces equalized odds from 33.30 to 28.18. This is an important negative-to-mixed result: the method is strongest when the fairness intervention can leverage synthetic balancing without overly distorting the target boundary, but on the hardest setting it still exchanges a small amount of utility for a meaningful fairness gain.

The right half of Table 1 reports data-scarce training on *Smiling/Male*. AIM-Fairremains the strongest method at 50% and 25% real-data usage, and yields the best worst-group accuracy and equalized odds even at 10%. The advantage over DiGA is especially clear in worst-group accuracy: 83.99 versus 81.10 at 50%, 82.54 versus 78.30 at 25%, and 81.73 versus 78.80 at 10%.

Table 3 compares AIM-Fairwith alternative training strategies. Several baselines achieve high average accuracy but remain much less fair. For example, in CelebA *Smiling/Male*, data supplementation attains 89.77 accuracy but only 73.67 worst-group accuracy and 21.96 equalized odds, whereas AIM-Fairreaches 84.20 worst-group accuracy and 6.07 equalized odds with 89.02 accuracy. Training only on synthetic data is fairer than the biased real baseline, but its average accuracy drops markedly, confirming that synthetic data alone are not enough. Full fine-tuning is competitive, yet AIM-Fairimproves on it in all three reported fairness metrics on CelebA *Smiling/Male* and on UTKFace.

The UTKFace results are similarly instructive. Data supplementation gives the highest average accuracy at 90.25, but fairness remains poor with 11.85 equalized odds. AIM-Fairattains 88.30 accuracy, 84.26 worst-group accuracy, and 4.81 equalized odds, improving both worst-group accuracy and equalized odds over full fine-tuning (83.00 and 5.40) while maintaining competitive accuracy.

5. Analysis

5.1. Why Selective Fine-Tuning Matters

Table 4 isolates the effect of the parameter-selection rule. Random selection and generic SubTuning can occasionally preserve accuracy, but their fairness is inconsistent. On CelebA *Smiling/Male*, for instance, updating one block with SubTuning obtains the best average accuracy of 90.04 but only 80.64 worst-group accuracy and 12.45 equalized odds. Our absolute-difference mask yields the best worst-group accuracy, equalized odds, and group-accuracy standard deviation in this setting. The same pattern broadly holds on *Smiling/Young* and UTKFace: architecture-only partial tun-

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Table 1. Comparisons to other methods on the CelebA dataset under settings of varied target and protected attributes.

Methods	T: Smiling, P: Male			T: Smiling, P: Young			T: Young, P: Male		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [6]	88.20	70.10	25.30	88.30	71.50	15.60	77.70	42.00	52.00
CVaR DRO [10]	87.30	74.00	22.80	87.00	76.10	13.90	75.40	42.30	48.80
EIIL [4]	87.90	75.60	19.70	87.90	72.50	13.30	77.50	45.60	39.20
LfF [15]	87.10	77.50	17.00	85.30	72.90	14.30	77.40	44.20	43.60
JTT [11]	88.00	74.80	19.40	87.60	73.30	14.20	76.30	43.60	47.70
MAPLE [25]	88.10	72.00	19.60	88.10	73.60	13.60	76.30	46.20	43.50
DiGA [24]	88.40	81.90	7.40	89.10	78.50	9.50	**80.00**	51.30	33.30
AIM-Fair (Ours)	**89.02**	**84.20**	**6.07**	**90.21**	**87.78**	**5.44**	78.19	**54.89**	**28.18**

Table 2. Comparisons to other methods on the CelebA dataset (T=Smiling, P=Male) under settings of training set sizes.

Methods	Samples Ratio = 50%			Samples Ratio = 25%			Samples Ratio = 10%		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [6]	87.50	67.80	26.10	87.10	65.90	27.70	86.90	62.80	28.90
CVaR DRO [10]	86.60	72.90	22.10	86.60	72.30	22.40	85.50	69.10	27.30
EIIL [4]	86.20	71.30	22.50	85.90	69.60	25.40	86.80	64.20	26.70
LfF [15]	86.90	75.50	19.40	85.90	72.10	23.60	85.50	66.10	27.70
JTT [11]	87.30	72.90	20.10	86.70	71.10	20.60	86.80	67.10	23.10
MAPLE [25]	87.40	73.70	23.80	87.00	72.70	24.20	85.60	69.20	27.10
DiGA [24]	88.40	81.10	7.80	88.40	78.30	8.00	**88.30**	78.80	8.40
AIM-Fair (Ours)	**88.85**	**83.99**	**6.28**	**88.89**	**82.54**	**7.59**	87.90	**81.73**	**8.16**

Table 3. Comparisons of varied training strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA: Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				UTKFace: Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Baseline [6]	T=0	83.57	98.50	89.23	71.52	23.84	10.65	78.72	96.09	90.19	78.72	17.37	6.92	79.58	96.16	88.86	79.58	16.59	7.26
	T=1	95.36	71.52					94.37	86.42					95.75	83.96				
Trained on Synthetic Data	T=0	86.82	89.40	85.98	78.52	7.87	4.07	85.45	85.76	85.12	82.42	**3.50**	**1.34**	82.71	88.12	85.44	82.71	5.50	**1.96**
	T=1	86.39	78.52					82.42	85.27					84.90	86.04				
Data Supplementation [19]	T=0	84.39	98.39	**89.77**	73.67	21.96	9.76	78.05	95.36	90.35	78.05	17.31	7.02	82.12	93.97	**90.25**	82.12	11.85	5.24
	T=1	95.40	73.67					94.90	87.62					95.62	89.28				
Data Repairing [5, 22]	T=0	82.82	98.10	89.72	75.05	21.18	9.50	79.50	95.19	**90.68**	79.50	15.69	6.30	81.15	94.22	89.42	81.15	13.37	5.89
	T=1	96.04	75.05					94.51	88.35					95.71	86.61				
Linear Prob	T=0	84.55	96.22	89.69	77.41	17.51	7.71	85.20	93.81	90.14	85.20	8.61	3.18	81.06	91.51	88.40	81.06	10.59	5.14
	T=1	94.82	77.41					90.43	87.88					94.59	86.42				
Fully Fine-Tuning	T=0	88.81	91.62	88.77	82.74	6.83	3.31	85.98	90.06	87.85	85.98	4.37	1.68	83.00	88.04	87.90	83.00	5.40	2.99
	T=1	89.55	82.74					86.02	86.81					90.74	89.83				
AIM-Fair (Ours)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	**4.81**	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

Table 4. Comparisons of varied partial fine-tuning strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA: Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				UTKFace: Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Best Random Selection	T=0	87.93	92.81	89.33	82.17	9.14	4.09	85.52	91.53	88.95	85.52	6.02	2.16	83.50	88.85	87.92	83.50	5.38	2.61
	T=1	91.31	82.17					88.25	87.63					90.30	89.01				
Best Sub-Tuning [9] (By Updating One Block)	T=0	87.50	94.75	**90.04**	80.64	12.45	5.52	87.28	94.63	**90.33**	87.28	7.34	3.00	83.05	89.56	**89.01**	83.05	6.51	3.56
	T=1	93.09	80.64					89.15	87.32					92.07	91.34				
Best Sub-Tuning [9] (By Freezing One Block)	T=0	88.69	91.09	89.06	83.48	7.00	3.00	86.55	91.89	88.65	86.55	**5.34**	2.25	83.58	88.34	87.42	83.58	**4.76**	**2.23**
	T=1	90.48	83.48					87.08	86.55					88.96	88.81				
Selective Fine-Tuning (Cosine Similarity)	T=0	86.07	91.77	88.85	83.33	8.40	3.61	87.43	92.96	90.26	87.43	5.53	**2.05**	83.72	88.82	88.21	83.72	5.10	2.64
	T=1	91.53	83.33					89.52	88.65					90.14	90.15				
AIM-Fair (Ours) (Absolute Difference)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	4.81	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

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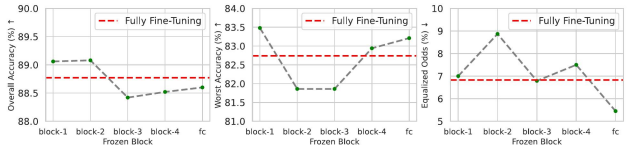

Figure 3. Layer-wise freezing on CelebA *Smiling/Male*. Freezing only the classifier head improves worst-group accuracy and equalized odds over full fine-tuning, showing that partial adaptation can help fairness, but heuristic block selection remains weaker than the mask used in AIM-Fair.

ing is competitive in utility, yet fairness-aware mask construction is more reliable for the full trade-off.

Figure 3 supports the same conclusion from a different angle. Freezing only the final fully connected layer already improves worst-group accuracy and equalized odds over full fine-tuning, suggesting that not all pretrained parameters should move when adapting with synthetic data. However, layer-wise heuristics alone do not identify the

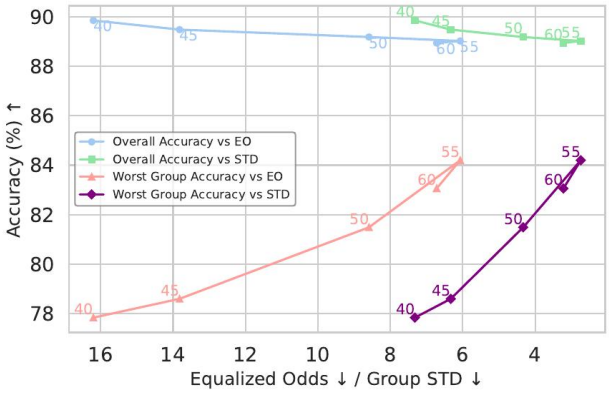

Figure 4. Effect of the top-k ranking threshold on CelebA *Smiling/Male*. Larger k generally improves worst-group accuracy and lowers fairness gaps, but too much unfreezing slightly reduces overall accuracy. The best balance occurs around $k = 55$.

Figure 5. Visual comparison of generated samples. A single plain prompt yields limited diversity, whereas contextual prompts cover richer variations in age cues, hairstyle, accessories, and head pose while preserving the desired target/protected attributes.

Table 5. Classification results on CelebA dataset (T=Smiling, P=Male) under settings of different prompt types and numbers.

Prompt Types (Number)	Targe	Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Plain Prompt (1)	t=0	65.18	89.44	71.65	43.28	34.20	17.07	83.10	77.72	72.75	59.53	9.30	8.96
	t=1	77.47	43.28					59.53	68.83				
Contextual Prompts (25)	t=0	77.12	89.25	81.85	69.73	16.3	7.65	76.72	67.90	79.80	67.90	8.82	8.75
	t=1	85.99	69.73					85.16	91.09				
Contextual Prompts (50)	t=0	89.47	89.80	82.66	72.22	**5.07**	7.66	87.72	90.43	**85.85**	79.67	4.36	4.20
	t=1	77.26	72.22					79.67	82.67				
Contextual + Head Poses (50)	t=0	86.82	89.40	**85.98**	**78.52**	7.87	**4.07**	85.45	85.76	85.12	**82.42**	**3.50**	**1.34**
	t=1	86.39	78.52					82.42	85.27				

best subset. Figure 4 then shows the effect of the mask size: increasing k from 40 to 55 steadily improves worst-group accuracy and reduces both equalized odds and group-accuracy variance, while overall accuracy decreases only slightly. Past this point the trade-off flattens, so the best setting uses $k = 55$.

These ablations also suggest that the benefit is not simply due to reducing adaptation capacity. If stronger regularization were the only mechanism, random selection or heuristic freezing would be expected to approach the same fairness levels once a comparable number of tensors is updated. The gap in Table 4 indicates that AIM-Fairbenefits from targeting fairness-relevant, domain-stable parameters rather than merely tuning fewer parameters.

5.2. Why Contextual Synthetic Data Matters

Table 5 and Figure 5 show that prompt design is a major determinant of downstream fairness. A single plain prompt performs poorly, with only 71.65 accuracy and 43.28 worst-group accuracy on CelebA *Smiling/Male*. Moving to 25 contextual prompts substantially improves both utility and fairness. Using 50 contextual prompts improves further, and adding explicit head-pose diversity produces the best overall trade-off in the main experiments. On *Smiling/Male*, contextual prompts with head poses raise worst-group accuracy to 78.52 while keeping equalized odds relatively low; on *Smiling/Young*, the same setting achieves the best worst-group accuracy, equalized odds, and group standard deviation. The qualitative samples in Figure 5 are consistent with this result: richer prompts reduce collapse toward a narrow facial prototype.

Table 6. Classification results on CelebA dataset (T=Smiling, P=Male) with different number of synthetic data.

Ratio To Real Data	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
0.5	t=0	88.65	93.96	**89.51**	80.80	10.31	4.90
	t=1	91.10	80.80				
1	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				
1.5	t=0	89.18	92.05	88.34	81.25	7.13	3.98
	t=1	88.38	81.25				
2	t=0	89.01	91.26	88.81	83.36	6.08	2.96
	t=1	91.26	83.44				

Table 7. Classification results on CelebA dataset (T=Smiling, P=Male) with varied bias ratio of the biased synthetic data.

Bias Ratio	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
Fully Fine-Tuning	t=0	88.81	91.62	88.77	82.74	6.83	3.31
	t=1	89.55	82.74				
4 : 6	t=0	85.72	91.71	**89.23**	83.90	8.78	3.77
	t=1	92.69	83.90				
3 : 7	t=0	87.05	91.62	88.90	83.52	7.72	3.26
	t=1	90.93	83.52				
2 : 8	t=0	88.01	91.76	88.97	83.16	7.30	3.28
	t=1	90.46	83.16				
1 : 9	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				

The amount and bias structure of synthetic data also matter. In Table 6, using balanced synthetic data at the same scale as the real training set gives the best fairness metrics, although 0.5× synthetic data yields slightly higher average accuracy. This suggests that too little synthetic data under-regularizes the spurious correlation, while more than 1× gives diminishing returns. Table 7 examines the biased synthetic set used for mask generation. A stronger skew of 1:9 produces the best worst-group accuracy and equalized odds, while milder bias ratios yield slightly higher average accuracy but weaker fairness. This supports the intended role of $\mathcal{D}_{S1}$: it should provide a clear contrast against balanced synthetic data so fairness-sensitive parameters can be identified more sharply.

5.3. When the Method Helps

Taken together, these ablations show that AIM-Fairworks best when synthetic supervision is both diverse and tightly targeted. Contextual prompts improve within-group cov-

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erage, the strongly biased $\mathcal{D}_{S1}$ makes fairness-sensitive tensors easier to identify, and a moderate mask size preserves most real-domain structure while still allowing debiasing updates. The same evidence also explains failure cases: weak prompts produce narrow prototypes, very small masks cannot move enough fairness-relevant parameters, and overly large masks begin to erode average accuracy by adapting too aggressively to the synthetic domain. This interpretation is also consistent with Table 3: methods that use synthetic data without controlling where the update lands can improve utility but still leave sizeable fairness gaps, whereas AIM-Fairturns the same intervention into more stable worst-group gains.

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6. Conclusion

We presented AIM-Fair, a fairness adaptation framework that pairs contextual synthetic data generation with parameter-selective fine-tuning. The core idea is simple: balanced synthetic data are most useful when they update only parameters that respond to fairness interventions without overreacting to synthetic-to-real domain shift. Across CelebA and UTKFace, this design consistently improves worst-group accuracy and equalized odds over full fine-tuning, generic partial tuning, and prior generative debiasing baselines. The analysis further shows that the gains come from both sides of the method: diverse contextual prompts improve the usefulness of synthetic data, and gradient-difference masking determines where that signal should enter the pretrained model. The harder Young/Male setting also highlights an important boundary case, where fairness improves but average accuracy does not, emphasizing that the method strengthens the fairness-utility trade-off rather than eliminating it.

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